

Crude Blend Optimizer

ML surrogate + differential evolution for crude-diet and FCC-severity planning at a 650,000 bbl/day refinery, built only on free public data

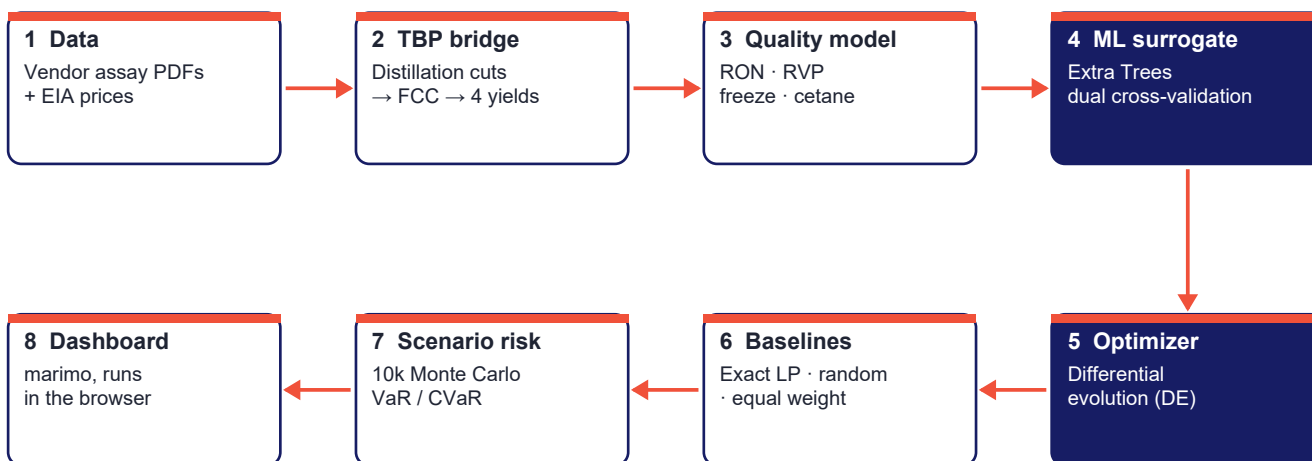
Repo: github.com/batestguy/dangote-refinery-optimizer · **Live demo:** batestguy.github.io/dangote-refinery-optimizer (first visit can take up to ~5 min to set up; later visits are fast)

In one sentence. Given five real crude oils and market prices, the system picks the blend and the cracking-unit severity that maximize refining margin per barrel while meeting fuel-quality specs. It then proves the answer is the true optimum and stress-tests it on 10,000 realistic price scenarios. The plant is sized like the Dangote Refinery, but **no company data is used**. Every number is computed from a published curve, cited, or explicitly labeled ASSUMED / PLACEHOLDER.

1 · The problem

Element	Definition
Decision	Blend shares $x \geq 0$ of 5 crudes ($\sum x = 1$) and FCC severity s between 0 and 1 (a normalized conversion proxy, not a reactor temperature)
Objective	Maximize margin/bbl = $\sum \text{product yield} \times \text{price} - \sum \text{crude share} \times \text{crude cost}$, over 4 product pools (gasoline, diesel, jet, petrochem)
Constraints	Blend API 30–45 · sulfur ≤ 1.5 wt% · gasoline RON ≥ 91 · RVP ≤ 60 kPa · jet freeze ≤ -47 °C · diesel cetane ≥ 45
Crudes	Bonny Light, Forcados, Qua Iboe (Nigeria); Alaska North Slope, Thunder Horse (US). Assays from TotalEnergies and ExxonMobil
Success gates (set before modeling)	Surrogate $R^2 > 0.90$ (random CV) · margin uplift vs naive blend $> 10\%$ · optimizer checked against an exact optimum

2 · Workflow



Navy boxes are the ML and optimization core; the others are data, physics, and delivery. Each stage in detail:

Stage	What it does	Technique	Why it matters
1 Data	Parses two vendor PDF assay formats into validated records (API, sulfur, full boiling curve); pulls EIA crude and product prices	PDF parsing, schema validation, cache-first API client	Real inputs; runs offline from committed artifacts
2 TBP bridge	Integrates each crude's boiling curve into cuts, routes them through a cited FCC and octane-unit model to 4 product yields	Numerical integration, mass balance, textbook constants (Gary & Handwerk)	Checked against the vendor's own published cut table: ± 0.2 vol%. Ground truth before any ML

Stage	What it does	Technique	Why it matters
3 Quality	Computes RON, RVP (index blending), jet freeze point, diesel cetane from the same streams	Standard refinery blending rules	Specs actually bind at the optimum, so the choice is non-trivial
4 Surrogate	Extra Trees regressor learns bridge yields from blend features; 4,000 Dirichlet-sampled training blends	Multi-output ETR, 150 trees, depth 14; random 5-fold and leave-one-crude-out CV	Fast stand-in for a slow process model; honest test on unseen crudes
5 Optimizer	Differential evolution over (x, s) with simplex repair and linear + quadratic constraint penalties	Vectorized DE (SciPy); whole population evaluated in one NumPy pass	127 s → 0.85 s per run; batch path pinned equal to scalar path by tests
6 Baselines	Compares DE against an equal-weight blend, best of 10k random blends, and an exact LP	LP reformulation: substitute $u = s \cdot x$ per crude (the physics are linear in severity)	Proves DE finds the true optimum instead of just reporting a number
7 Risk	10,000 scenarios from real 2015–2025 EIA monthly price changes; LP re-solved per scenario	12-month block bootstrap (keeps correlation and seasonality), VaR/CVaR, tornado	Shows how the plan behaves when markets move, not only at base prices
8 Delivery	Interactive dashboard; the same file runs locally, on a server, or fully in the browser	marimo + Pyodide (WebAssembly) on GitHub Pages; CI with 140 offline tests	Anyone can run the optimizer from a link, including on a phone

3 · Headline results

Question	Result
Best diet at base prices	~100% Alaska North Slope at maximum FCC severity: its sour-crude discount outweighs the cost of meeting specs
Margin at the optimum	\$15.67/bbl (exact LP); DE matches it to within \$0.005/bbl
Uplift vs naive equal-weight blend	+10.0% (\$15.67 vs \$14.23/bbl) on the exact model; +11.3% through the surrogate (its ~1% error)
Surrogate accuracy	Random 5-fold $R^2 \geq \mathbf{0.982}$ on all four products
Value of re-optimizing as prices move	+\$2.56/bbl on average; worst-5% tail (CVaR) improves from −\$3.50 to +\$1.42
Downside risk (10k scenarios)	VaR 5% \$6.25 · CVaR 5% \$1.42 · probability of loss 0.8%
Does the best diet change?	Yes, two regimes: Alaska North Slope optimal in 50.5% of scenarios, Forcados in 47.9%
Robustness	100 DE seeds: margin $\sigma = \$0.0015/\text{bbl}$; everything seeded (20260919)
Runtime	DE ≈ 0.85 s (exact model), ≈ 3.4 s (surrogate); 10k-scenario Monte Carlo $\approx 85\text{--}105$ s

4 · The ML part, for an ML reader

Setup. Features are blend-weighted crude properties (API, sulfur), blend-weighted boiling-cut fractions, and severity. Labels are the bridge's four pool yields. They are generated by a cited physical model, **not plant measurements**. Two cross-validations are always reported side by side:

Target	Random 5-fold R^2	Leave-one-crude-out R^2	Reading
Gasoline	0.989	0.935	Generalizes well
Diesel	0.997	0.781	Moderate drop on an unseen crude
Jet	0.982	0.114	Held-out Alaska North Slope has 12.6% kerosene vs 14–17% in training. Trees cannot extrapolate
Petrochem	0.998	0.969	Generalizes well

Random CV measures interpolation, which is how the optimizer uses the model (it only blends crudes it was trained on). Leave-one-crude-out measures extrapolation to a new crude. Reporting only the first would overstate the model; the jet result is disclosed on the dashboard itself.

An honest caveat worth knowing. Because the yield physics turned out to be linear in the decision variables, an exact LP solves this version of the problem, and DE simply ties it. In this version the surrogate and DE do not beat classical optimization; they are the scaffolding for the realistic case, where the process model is nonlinear or is a slow rigorous simulator (or plant data), and an LP is no longer exact. The project is built so that the same LP comparison will show whether DE pulls ahead then.

5 · Why this matters to recruiters

What hiring teams look for	Evidence in this project
End-to-end ownership	Raw PDFs and APIs → physics model → ML → optimization → risk analysis → deployed app, one person
Honest evaluation	Dual CV, mandatory baselines, success gates fixed before modeling, weak jet score published instead of hidden
Beyond model.fit()	Recognized the problem's linear structure and built an exact LP to verify the ML/optimizer result
Domain grounding	Refinery engineering cited to standard texts; parser validated against published assay tables
Decision science	Correlated Monte Carlo, VaR/CVaR, sensitivity (tornado), value-of-re-optimization quantified in \$/bbl
Engineering quality	140 offline tests, CI on every push, locked dependencies, seeded runs, vectorized code 150× faster
Traceability	Every constant cited or labeled ASSUMED/PLACEHOLDER with a refresh path (audit trail in docs/methodology.md)
Communication and delivery	Dashboard that explains itself section by section; free hosting; works on phones

6 · Shortcomings, and how the company could help

Shortcoming	Effect	What company support would fix it
No plant data: the surrogate learns a cited model, not the real refinery	Accuracy against actual operations is unknown	Historian and lab (LIMS) data on crude runs and unit yields, to train and validate on real outcomes
FCC simplified: constant product split, linear severity response, severity is a proxy	Misses real feed-quality and nonlinear effects	FCC test-run / severity-sweep data, or access to a rigorous simulator to generate realistic labels
Crude price differentials ASSUMED; petrochem price is a PLACEHOLDER	Margin levels, and possibly the chosen diet, could shift	Procurement's actual crude purchase prices and product netbacks
Only 5 crudes with open assays (Arab Light, Urals unavailable)	Slate differs from what a real plant buys	The company's assay library for crudes actually purchased
Surrogate cannot extrapolate to a new crude (jet R^2 0.11 out-of-sample)	Adding a new grade needs retraining	More crudes in training, physics-informed features; company assays widen coverage
Single period, USD price-taker; no inventory, logistics, unit capacities, or FX	Not yet a full planning tool	Planning constraints (tankage, cargo sizes, unit limits, Naira exposure) for a multi-period model
ML/DE ties the LP on this linear physics	Value over classical LP not yet shown	Benchmark against the company's existing planning LP on nonlinear, plant-calibrated models
Browser demo: first visit up to ~5 min	Some visitors leave early	Internal server hosting makes it load instantly

7 · Suggested path with company support

Phase	Work	Outcome
1 · Calibrate	Refit the bridge and surrogate on historical crude runs, unit yields and lab qualities	Model error measured against real operations
2 · Shadow run	Run alongside the existing planning process each month, without acting on it	Evidence of \$/bbl difference vs current plans
3 · Extend	Nonlinear blending, crude-specific FCC behavior, multi-period inventory and FX	A model where ML and DE can add value beyond an LP
4 · Pilot	Decision-support dashboard for the planning team, hosted internally	Faster what-if analysis and risk-aware diet choices

8 · Glossary (refinery terms)

Term	Meaning
Crude assay	Lab report describing a crude oil: density (API gravity), sulfur, and its boiling curve
TBP curve	True boiling point curve: % of the crude that boils off by each temperature; cutting it gives naphtha, kerosene, diesel, etc.
FCC / severity	Fluid catalytic cracker converts heavy gas oil into gasoline and lighter products; higher severity means more conversion
RON / RVP	Gasoline octane (knock resistance) and vapor pressure (volatility); both have legal specs
Cetane / freeze point	Diesel ignition quality; lowest temperature at which jet fuel stays liquid
Margin/bbl	Product revenue minus crude cost per barrel processed (operating costs excluded here)
LP	Linear program: exact optimizer when objective and constraints are linear; the standard refinery planning tool
VaR / CVaR 5%	Margin in the worst 5% cutoff scenario / average margin across the worst 5% of scenarios

Stack: Python 3.12, NumPy, pandas, SciPy, scikit-learn, Plotly, marimo, Pyodide; uv for reproducible environments; GitHub Actions CI. Full methodology and data provenance are in the repository.